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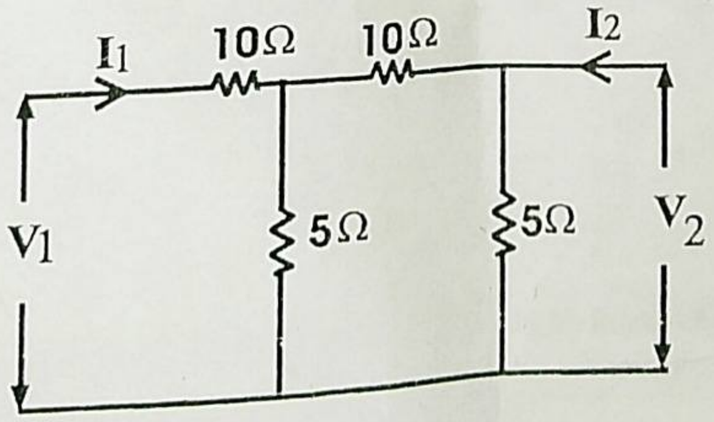
OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE, AI&ML) III - Semester (MAIN) Examinations Dec-2024/Jan 2025

BASIC ELECTRONICS

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer* ***Question No. 1*** *is compulsory and answer any* ***four questions*** *from the remaining questions (2- 7), each Question carries 12 Marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
iii) ***Missing data, if any, may suitably be assumed.***

			Marks	BT	CO
1.	a)	Write the differences between extrinsic and intrinsic semiconductors.	2	1	1
	b)	What is Stability factor of BJT.	2	1	2
	c)	What are the advantages of Negative feedback.	2	2	3
	d)	What are the characteristics of ideal op-amp.	2	1	4
	e)	Derive relation between FET parameters.	2	1	5
	f)	Compare LED and LCD.	2	3	2
2.	a)	Draw a neat circuit for Bridge rectifier with Choke filter. Derive I_{dc}, V_{dc}, I_{rms}, Ripple Factor and PIV.	6	4	1
	b)	Explain the construction and working of CRO.	6	3	1
3.	a)	Derive the h parameters of the given circuit	6	3	2



	b)	Derive Z_o, Z_i, A_V, A_i for CS amplifier.	6	3	2
4.	a)	Derive the general equation for LC oscillator and from that deduce the frequency of oscillation of Hartley Oscillator.	6	3	3
	b)	How the bandwidth of negative feedback amplifier increases, derive with the equations.	6	4	3

5. a)	Explain the construction and working of LVDT. Mention its applications.	6		
b)	Classify Photo Electric Devices and their working principle.	6	4	
6. a)	Explain the working principle and its usage as an integrator.	7	4	
b)	Design a circuit for which output is $V_0=2V_1-2V_2-5V_3$ using op-amps.	5	2	
7.	Write short notes on			
a)	FET	5	2	
b)	RC oscillators	4	2	
c)	Zener Regulator	3	3	

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